



Scanner field card

The barcodes you scan often, where every other setting lives in Zebra's manuals, what the beeps mean, and what to do when it stops working. Scan straight off this screen.

TWO MANUALS, AND WHICH ONE YOU NEED

MANUAL	SIZE	WHAT'S IN IT
QSG Quick Start Guide	22 pp	The everyday barcodes: pairing, factory reset, Enter and Tab keys. Also the paper copy in the box.
PRG Product Reference Guide	476 pp	Everything else. Every parameter the scanner has, with its barcode. Batch mode lives here, not in the short one.

Pages are given as **PDF page / printed section**, because the two differ throughout. **PDF 92 / 5-34** means page 92 of the file, printed as 5-34.

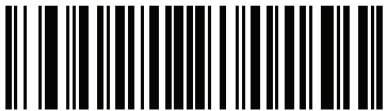
THE FOUR YOU'LL ACTUALLY SCAN

Zoom the page up with **Ctrl** and **+**, take the scanner out of the cradle, hold about 15 cm back and tilt slightly off the glare. Listen rather than watch:
high/low/high/low means the setting took.

START HERE

HID Bluetooth Classic

Keyboard mode, and discoverable. In this mode it types into whatever holds the cursor, which is what Sheets, Notes and Shopify POS all need. Scan it, then pair from **Settings** > **Bluetooth** straight away, the window is short.



HID BLUETOOTH CLASSIC

WIPES EVERYTHING

Set Factory Defaults

Resets every parameter, including your Enter suffix, any batch settings and the host mode. Only for recovering an unknown state, and always follow it with HID Bluetooth Classic. It does **not** unpair the tablet, so forget the device in Android Bluetooth first. Zebra's printed guide calls this *Restore Default Parameters*.



SET FACTORY DEFAULTS

KILLS TEXT INPUT

Symbol Native API

Application Mode. The scanner stops typing and speaks Zebra's own protocol, so only their Scan-To-Connect app can hear it. Sheets, Notes and every search bar receive **nothing at all, silently**. This is the barcode behind a scanner that reads fine but types nowhere.



SYMBOL NATIVE API

WIRED ONLY

USB Keyboard (HID)

Same keyboard behaviour over the cradle's USB cable. Shopify calls wired the most reliable way to run this scanner, so bench work can go through the cradle and only the aisles need Bluetooth. Doesn't affect Bluetooth behaviour either way.



USB KEYBOARD (HID)

MAKING IT JUMP TO THE NEXT CELL

Out of the box the scanner types the barcode and stops, so every scan lands on top of the last one in the same cell. Two parameters control this, and only one of them needs touching.

PARAMETER	FACTORY SETTING	WHAT TO DO
Suffix value #98, #104	7013 <CR><LF>	Nothing. It is already Enter.
Transmission format #235	Data As Is	Scan <DATA> <SUFFIX 1> — PRG 92 / 5-34

Enter moves down, Tab moves right. Enter drops the cursor a row, so you capture straight down one column. Tab moves a column right, so you can fill barcode, quantity, location across a row. Pick one, not both. The Tab Key barcode is on **qsg** page 10.

Test: tap a cell, scan two things. Two rows means it took. Both in one cell means it didn't.

Pairing alone does nothing here. Three separate things must be true before a scan reaches the cart.

01 Allow HID mode is on.

Shopify POS > ... menu > Settings > *Allow HID mode*. Without it POS ignores keyboard input entirely, and the scanner will type fine into Notes while doing nothing in POS.

02 Android permissions are right.

Nearby devices *Allow*, Location *while using the app*, Camera *allow*, and *Pause app activity if unused* switched **off**. That last one puts the app to sleep and makes the scanner "randomly stop working" days later.

03 The product has a Barcode.

POS matches the variant's *Barcode* field, not the SKU field. An empty Barcode field means "Unrecognized barcode" no matter how perfect the scanner setup is.

Never tap the search bar first. Scanning with search focused makes POS search instead of adding to the cart, and on Android there is an unresolved bug where scans stop registering entirely once you have touched it. Scan from the home screen.

Proving the chain works: scan any random household item in POS. "*Unrecognized barcode*" is the result you want. It means the whole path is healthy and only your catalogue data is missing.

TO CHANGE THIS	MANUAL	PAGE
Pair over Bluetooth — HID Bluetooth Classic	QSG	11
Factory reset , Tab key, Enter key, Caps Lock	QSG	10
Turn the Enter suffix on — transmission format #235	PRG	92 / 5-34
Change what the prefix or suffix is #98, #104	PRG	91 / 5-33
Cancel a stuck programming sequence	PRG	414 / G-3
Numeric barcodes, for typing values into settings	PRG	412-414 / App. G
Batch mode , all five versions #544	PRG	129-130 / 6-32
Send Batch Data, Persistent Batch Storage #1399	PRG	131 / 6-34
Auto-reconnect behaviour #604	PRG	122 / 6-25
Beeper volume #140	PRG	65 / 5-7
Beeper tone and duration	PRG	66 / 5-8
Trigger mode, and hands-free presentation mode #138, #630	PRG	74-75 / 5-16
Battery preservation #1765	PRG	73 / 5-15
Beep and LED meanings , scanner and cradle	PRG	39-43 / 3-1
Zebra's own troubleshooting table	PRG	51-52 / 4-5
Bluetooth range, linking and batch operation	PRG	134 / 6-37
Enable or disable barcode types	PRG	Chapter 13
Keyboard country layout	PRG	App. B
Configure from a PC instead — 123Scan	PRG	35 / Chapter 2

The shortcut for anything complicated: 123Scan is Zebra's free Windows tool. It walks you through settings in a wizard and prints the result as a *single* programming barcode, so you never hunt a 476 page

WHAT YOU CAN CHANGE

How it sounds Volume, tone, duration. Down for a quiet shop, up for a noisy garage. PRG 5-7	How it fires Trigger press, or hands-free presentation mode where it reads anything put in front of it. PRG 5-16	What it appends A prefix and up to two suffixes. Enter, Tab, or any ASCII value. PRG 5-33
When it sends Live on every scan, or buffered while out of range and flushed on reconnect. PRG 6-32	What it accepts Every symbology on or off individually, plus length limits per type. PRG ch.13	Keyboard layout Country layout, so letters and symbols aren't garbled. PRG App. B
Battery behaviour Preservation mode disconnects the battery after nine idle hours. PRG 5-15	Reconnect behaviour What it does when it walks out of range and comes back. PRG 6-25	

READING THE BEEPS

SOUND	LED	MEANING
Medium beep	● Green	Decoded. Normal, good scan.
High/low/high/low	● Green	Setting took. Successful program exit with a change.
Low, high	● Green	Bluetooth connected. The sound you want after pairing.
High, low	● Red	Bluetooth disconnected. It had a link and lost it.
Long low, long high	● Red	Page timeout, meaning no host in range, <i>or</i> a programming input error, <i>or</i> Cancel was scanned. Depends what you pointed it at.
4 long low	● Red	Transmission error, data ignored. Read it, had nowhere to send it.
Low, high, low, high	● Red	Batch memory full. New scans are being lost.
4 short sharp	● Red	Low battery, sounded on trigger release.
Low/medium/high	● Green	Powering on.

WHEN IT STOPS WORKING

Three different faults look similar and have completely different fixes. Find your symptom first, because reaching for a factory reset when you needed a Cancel costs you the pairing and the Enter suffix for nothing.

SYMPTOM A

Types fine in Notes, does nothing in Shopify POS

Not a scanner fault at all. The scanner is healthy. Check, in this order: **Allow HID mode** is on in POS settings; you did **not** tap the search bar before scanning; and the product actually has its **Barcode field** filled in Shopify admin. See the POS section above.

Config barcodes keep erroring — long low, long high, red

The scanner is stuck part way through a programming sequence and rejecting everything, because that error leaves it *in* program mode. Usually caused by catching a neighbouring barcode, since the config pages pack them close together.

01 **Scan Cancel.**

PRG page 414 / G-3. This exits cleanly and keeps your pairing and settings.

02 **Retry the setting, covering the neighbours with paper.**

Listen for high/low/high/low.

Cancel itself beeps red. "Cancel scanned" is listed under that same error beep, so a successful cancel sounds exactly like a failure. Don't keep re-scanning it. Do **not** factory reset for this fault.

Reads barcodes but nothing types anywhere, red LED

It decoded fine and had no valid host to deliver to. Either it is in Application Mode, or the Bluetooth link is gone. This is the full rebuild, and it costs you your settings, so only run it if Notes stays empty.

01 Force stop the Zebra app.

Scan-To-Connect holds the scanner in Application Mode. Nothing else can have it while that app runs.

02 Unpair in Android Bluetooth.

Gear icon next to DS2278, then Unpair. A stale record blocks the new pairing.

03 Scan Set Factory Defaults.

Clears whatever host mode it was left in.

04 Scan HID Bluetooth Classic.

Listen for high/low/high/low. It is not discoverable until you do this.

05 Pair from Android Bluetooth, immediately.

The window is short. Listen for low/high, that's the link established.

06 Test in Notes.

Text must appear. If it doesn't, no app setting will rescue it, so go back to step one.

07 Redo the Enter suffix.

The reset wiped it. `<DATA> <SUFFIX 1>` , `PRG` page 92.

08 Redo your batch settings, if you use them.

The reset wiped those too, including Persistent Batch Storage.

BATCH MODE, FOR LEAVING THE TABLET BEHIND

The scanner accepts most of the batch parameters but only one variant has actually been seen to work. Everything below is what was verified on the bench rather than what the manual promises, because the two don't fully agree.

BARCODE	RESULT	NOTES
Standard Batch Mode (02h)	● Accepted	Sets the mode. Must go green before the commands below mean anything.
Enter Batch Mode	● Accepted	Only valid once the mode is Standard. Scanned first, it errors.
Out of Range Batch Mode (01h)	● Accepted	The automatic alternative. Flushes on reconnect with no command.
Persistent Batch Enable (1)	● Accepted	Storage preference. Survives a power down.
Send Batch Data	● Refused	Still unresolved. Errors even with the mode set, and scans continue to transmit live rather than buffer.

Manual send does not work, and it isn't understood why. With Standard Batch Mode and Enter Batch Mode both accepted, scans still transmit live instead of buffering, and *Send Batch Data* then errors because there is nothing held. Nothing in Zebra's documentation gates it. Resolve this with **123Scan** over the USB cable, which reads the scanner's real parameter values back, rather than by inferring state from beeps.

What does work is the automatic route. Two barcodes, scanned once: *Out of Range Batch Mode* and *Persistent Batch Enable*, on **PRG** pages 130 and 131. Persistent matters because without it a flat battery loses everything sitting in the buffer.

The flush is fire and forget, and it will destroy your run if you get this wrong. On reconnect the scanner types the whole buffer as keystrokes and clears its memory. It has no idea whether anything received them. Reconnect with the tablet locked, or on the home screen, or with the sheet open but no cell selected, and every scan is gone. Verified the hard way: nothing arrives and there is no second copy. Persistent Batch Storage does not protect you here, it only survives a power cut.

Since nothing can trigger the flush by hand, you control it one of two ways. Both work. Pick one and stay with it.

Option A — park the tablet properly, never touch Bluetooth

Set it up once and the automatic flush always has somewhere correct to land, so you can forget about it entirely.

01 Screen timeout set to Never.

Settings > Display > Screen timeout. This is the one that catches people out: a locked screen has no focused cell, so a dump onto it dies.

02 Sheet open, starting cell tapped, tablet face up and plugged in.

Screen-on-forever drains the battery, and Do Not Disturb keeps notifications from covering it.

03 Don't use the tablet for anything else during the run.

Switching apps moves the focus, and the next flush follows the focus.

Parking the tablet at the end of the aisle you're working, rather than back at the bench, means most scans go in live and the buffer only catches the strays. Less riding on the flush.

Option B — drive it with the Bluetooth switch

More deliberate, and immune to the tablet's state. Put the Bluetooth tile in your quick settings so it's a swipe and a tap.

01 Bluetooth off on the tablet before you set off.

The scanner now cannot reconnect while you aren't looking.

02 Scan the warehouse.

Everything buffers. Scans beep and flash green exactly as if they were sending, so don't read that as confirmation.

03 Open the sheet and tap the starting cell, Bluetooth still off.

Nothing can flush yet.

04 Bluetooth on.

It reconnects and dumps where you chose. Scan *Normal* (00h) when you want live scanning back.

Never enable Auto-Reconnect on Bar Code Data alongside this. Zebra's explicit warning: the data batches and the scanner can then never reconnect. That's a permanent hang, not a glitch. **PRG** 122 / 6-25.

You cannot hear the difference between storing and sending. A buffered scan gives the same beep and green flash as a live one. The only distinct sound is low/high/low/high with red, meaning the memory is full and new scans are being discarded. Capacity is 30,720 bytes divided by characters plus three, roughly **1,900 EAN-13 scans**.

Bench test before the first real run. Store five scans, come back, and check two things: that all five arrive, and that they land on five rows rather than piling into one cell. The Enter suffix should still fire on replay, but prove it with five rather than discovering it after two hundred.

Red never means bad barcode. It means read fine, delivered nowhere. Always a host or link problem, never the label.

The suffix value was always Enter. Factory default is `7013 <CR><LF>`. Only the transmission format needs moving off "Data As Is".

A factory reset costs you three things. The Enter suffix, the Bluetooth host mode, and any batch settings. Redo all of them afterwards.

Never pair Out of Range Batch with Auto-Reconnect on Bar Code Data. Zebra's own warning: the data batches and the scanner can never reconnect.

Config barcodes sit close together. At screen zoom the scanner's field covers more than one. Cover the neighbours with paper.

Range is about 10 metres, less with shelving. It's a Class 2 radio. Out of range means scans are silently discarded unless batch mode is on.

Settings live on the scanner, not the tablet. Reprogramming it from another device changes it everywhere.

Barcode images are Shopify's published configuration barcodes for the Zebra DS2278. Page references are to Zebra's DS2278 Quick Start Guide (22 pp) and Product Reference Guide (476 pp), both saved in the project docs folder.